



Problem-Solution Fit

Date	03 July 2026
Team ID	
Project Name	Credit Card Approval Prediction
Maximum Marks	5 Marks

Problem-Solution Fit Canvas:

Use the framework below to ensure your proposed solution accurately addresses the root causes, behaviors, and limitations of your target audience. Fill out the canvas in the respective boxes just like the original visual template.

1. CUSTOMER SEGMENT(S) [CS] 1.Banks and Financial Institutions 2.Credit Card Analysts 3. Loan Officers 4. Customers applying for credit cards	6. CUSTOMER LIMITATIONS EG. BUDGET, DEVICES [CL] 1.Limited access to advanced AI systems. 2.Budget constraints. 3.Limited technical expertise. 4. Security and privacy concerns.	5. AVAILABLE SOLUTIONS PROS & CONS [AS] Manual Verification 1.Pros: Human judgment, flexible. 2.Cons: Slow, costly, error-prone. Basic Rule-Based Systems 1.Pros: Fast. 2.Cons: Less accurate, cannot learn from historical data.
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**2. PROBLEMS / PAINS
+ ITS FREQUENCY [PR]**

1. Manual verification takes a long time.
2. High number of credit card applications daily.
3. Human errors in approval decisions.
4. Delayed approval/rejection process.

**9. PROBLEM ROOT /
CAUSE [RC]**

1. Manual processing of applications.
2. Large amount of applicant data.
3. Difficulty in identifying high-risk applicants.
4. Lack of automated prediction systems.

**7. BEHAVIOR
+ ITS INTENSITY [BE]**

1. Customers expect instant approval status.
2. Banks require accurate risk assessment.
3. High dependence on online banking services.

**3. TRIGGERS TO ACT
[TR]**

1. Large volume of applications.
2. Need for faster and more accurate decisions.
3. Increasing demand for digital banking services.
4. Requirement to reduce operational costs.

**10. YOUR SOLUTION
[SL]**

Develop an **AI-based Credit Card Approval Prediction System** using **Logistic Regression, Decision Tree, Random Forest, and XGBoost**. The best-performing model is integrated with a **Flask web application** and deployed using **IBM Watson Machine Learning** to provide secure, fast, and accurate real-time credit card approval predictions.

**8. CHANNELS OF
BEHAVIOR [CH]**

ONLINE

1. Bank websites
2. Mobile banking apps
3. Online credit card portals

OFFLINE

1. Bank branches
2. Customer service centers

**4. EMOTIONS
BEFORE / AFTER [EM]**

Before:

1. Uncertainty
2. Frustration
3. Long waiting time

After:

1. Quick response
2. Confidence in decision
3. Better customer satisfaction